

CS 451: Computer Graphics and Game Design

Course Code: CS 451

Course Name: Computer Graphics and Game Design

L-T-P-C: 2-0-2-3

Prerequisites: Experience in writing computer-code.

Intended for: B.Tech. (all branches) II/III/IV year

Distribution: Elective for B.Tech. II/III/IV year

Approval: 28th Senate

Course Contents

- **Introduction:** Motivation; Types of games, Different aspects of game design; Different components in a game; Game engines; Geometric primitives, 2D and 3D linear transforms, Homogeneous matrices; Examples of games. [5 Lectures]
- **Sprites and animation:** Different image formats; Polygon file formats; Creating sprites; Rigging; Animations using sprite-sheets; Animations using keyframes; Animation controllers. [6 Lectures]
- **Level design:** Scenes; Tiles, visual continuity in tiles; Adding objects to scene; Prefabs; Lighting, RGB space, transparency, texture mapping; Collectibles; Navigation and pathfinding. [6 Lectures]
- **World interaction:** Physics engines; Gravity simulation; Rigid body interaction; Collisions. [6 Lectures]
- **User interface:** Layout; Menu system; Visual components; Event system; Skins. [3 Lectures]
- **Audio:** Audio assets; Different audio formats; Audio mixing. [2 Lectures]

Lab Exercises

Lab to be conducted on a 2-hour slot, in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course shall be appropriately sequenced for synchronization with the laboratory. The students will progressively design and write a 2D video game as part of the lab.

- Lab1-2: Installation of a game engine, e.g., Unity, familiarization of the GUI. Conceptualize the theme for a 2D game.
- Lab3-4: Character design, sprites, movement and character control.
- Lab5-7: Level design: design of the world in form of tiles along with interactive and collectible objects.
- Lab8-9: Design of interaction between the player and the world, optionally using the physics engine.
- Lab9-11: Design of menus and user interaction in mobile platform.
- Lab12: Insert audio.

Textbooks:

1. Nystrom Robert, **Game Programming Patterns**, 3rd Edition, Genever Benning, 2014



Reference Books:

1. Paris Buttfield-Addison et al., **Unity Game Development Cookbook: Essentials for Every Game**, O'Reilly Media, 2019.